

Q3 Constant acceleration equations

Name: _____

Class: _____

Date: _____

Time: **415 minutes**

Marks: **348 marks**

Comments:

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Q1.

A box, of mass 3 kg, is placed on a slope inclined at an angle of 30° to the horizontal. The box slides down the slope. Assume that air resistance can be ignored.

- (a) A simple model assumes that the slope is smooth.
- (i) Draw a diagram to show the forces acting on the box. (1)
- (ii) Show that the acceleration of the box is 4.9 m s^{-2} . (2)
- (b) A revised model assumes that the slope is rough. The box slides down the slope from rest, travelling 5 metres in 2 seconds.
- (i) Show that the acceleration of the box is 2.5 m s^{-2} . (2)
- (ii) Find the magnitude of the friction force acting on the box. (3)
- (iii) Find the coefficient of friction between the box and the slope. (5)
- (iv) In reality, air resistance affects the motion of the box. Explain how its acceleration would change if you took this into account. (2)
- (Total 15 marks)**

Q2.

A boat is initially at the origin, heading due east at 5 m s^{-1} . It then experiences a constant acceleration of $(-0.2\mathbf{i} + 0.25\mathbf{j}) \text{ m s}^{-2}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) State the initial velocity of the boat as a vector. (1)
- (b) Find an expression for the velocity of the boat t seconds after it has started to accelerate. (2)
- (c) Find the value of t when the boat is travelling due north. (3)
- (d) Find the bearing of the boat from the origin when the boat is travelling due north. (6)
- (Total 12 marks)**

Q3.

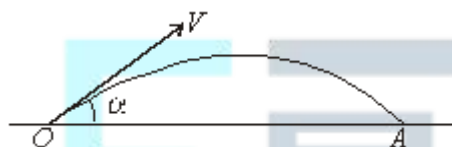
A girl throws a ball vertically upwards with a speed of 10.5 m s^{-1} and subsequently catches it at the same point from which it was thrown.

Find:

- (a) the greatest height that the ball reaches above the point from which it is thrown; (2)
 - (b) (i) the time that the ball takes to reach the greatest height; (2)
 - (ii) the time between the ball being thrown and being caught. (1)
- (Total 5 marks)**

Q4.

A golf ball is projected from a point O with initial velocity V at an angle α to the horizontal. The ball first hits the ground at a point which is at the same horizontal level as O , as shown in the diagram.

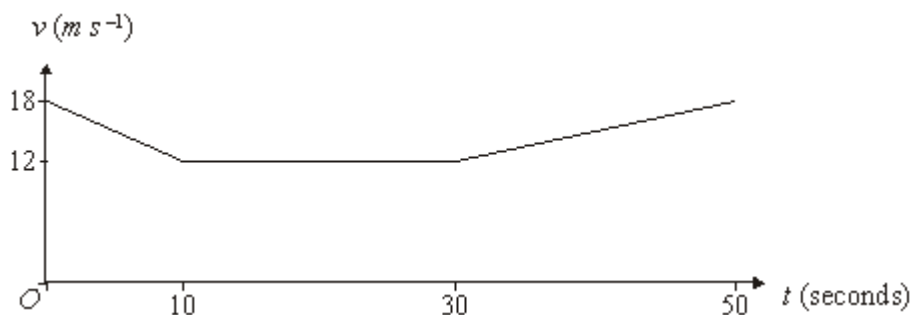


It is given that $V \cos \alpha = 6u$ and $V \sin \alpha = 2.5u$.

- (a) Show that the time taken for the ball to travel from O to A is $\frac{5u}{g}$. (4)
 - (b) Find, in terms of g and u , the distance OA . (2)
 - (c) Find V in terms of u . (2)
 - (d) State, in terms of u , the least speed of the ball during its flight from O to A . (1)
- (Total 9 marks)**

Q5.

The velocity-time graph below represents the three stages of the motion of a coach moving along a straight horizontal road. Initially the coach has velocity 18 m s^{-1} .



- (a) During the first stage of the motion, the coach decelerates at a constant rate of $a \text{ m s}^{-2}$ for 10 seconds until it reaches a velocity of 12 m s^{-1} .

(i) Find the value of a .

(2)

(ii) Find the distance that the coach travels during the 10 seconds.

(2)

- (b) During the second stage of the motion, the coach travels for 20 seconds with constant velocity 12 m s^{-1} . Find the distance that the coach travels during these 20 seconds.

(1)

- (c) During the third stage of the motion, the coach travels with constant acceleration, reaching a velocity of 18 m s^{-1} after a further 20 seconds.

Find the average speed of the coach during the 50 seconds of the motion.

(4)

(Total 9 marks)

Q6.

A particle P moves with acceleration $(-3\mathbf{i} + 12\mathbf{j}) \text{ m s}^{-2}$. Initially the velocity of P is $4\mathbf{i} \text{ m s}^{-1}$.

- (a) Find the velocity of P at time t seconds.

(2)

- (b) Find the speed of P when $t = 0.5$.

(3)

(Total 5 marks)

Q7.

- (a) A small stone is dropped from a height of 25 metres above the ground.

(i) Find the time taken for the stone to reach the ground.

(2)

(ii) Find the speed of the stone as it reaches the ground.

(2)

- (b) A large package is dropped from the same height as the stone. Explain briefly why the time taken for the package to reach the ground is likely to be different from that for the stone.

(2)

(Total 6 marks)

Q8.

A van moves from rest on a straight horizontal road.

- (a) In a simple model, the first 30 seconds of the motion are represented by three separate stages, each lasting 10 seconds and each with a constant acceleration.

During the first stage, the van accelerates from rest to a velocity of 4 m s^{-1} .

During the second stage, the van accelerates from 4 m s^{-1} to 12 m s^{-1} .

During the third stage, the van accelerates from 12 m s^{-1} to 16 m s^{-1} .

- (i) Sketch a velocity–time graph to represent the motion of the van during the first 30 seconds of its motion.

(3)

- (ii) Find the total distance that the van travels during the 30 seconds.

(4)

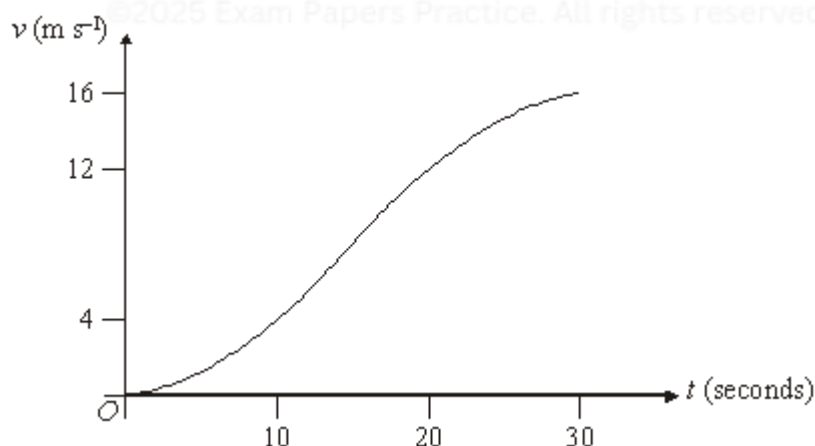
- (iii) Find the average speed of the van during the 30 seconds.

(2)

- (iv) Find the greatest acceleration of the van during the 30 seconds.

(2)

- (b) In another model of the 30 seconds of the motion, the acceleration of the van is assumed to vary during the first and third stages of the motion, but to be constant during the second stage, as shown in the velocity–time graph below.



The velocity of the van takes the same values at the beginning and the end of each stage of the motion as in part (a).

- (i) State, with a reason, whether the distance travelled by the van during the first

10 seconds of the motion in **this** model is greater or less than the distance travelled during the same time interval in the model in part (a).

(2)

- (ii) Give one reason why **this** model represents the motion of the van more realistically than the model in part (a).

(1)

(Total 14 marks)

Q9.

A small stone is dropped from a high bridge and falls vertically.

- (a) Find the distance that the stone falls during the first 4 seconds of its motion.

(3)

- (b) Find the speed of the stone when it has been falling for 4 seconds.

(2)

(Total 5 marks)

Q10.

A car travels along a straight horizontal road. The motion of the car can be modelled as three separate stages.

During the first stage, the car accelerates uniformly from rest to a velocity of 10 m s^{-1} in 6 seconds.

During the second stage, the car travels with a constant velocity of 10 m s^{-1} for a further 4 seconds.

During the third stage of the motion, the car travels with a uniform retardation of magnitude 0.8 m s^{-2} until it comes to rest.

- (a) Show that the time taken for the **third** stage of the motion is 12.5 seconds.

(2)

- (b) Sketch a velocity–time graph for the car during the three stages of the motion.

(4)

- (c) Find the total distance travelled by the car during the motion.

(3)

(Total 9 marks)

Q11.

The points A and B have position vectors $(3\mathbf{i} + 2\mathbf{j})$ metres and $(6\mathbf{i} - 4\mathbf{j})$ metres respectively. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in a horizontal plane.

- (a) A particle moves from A to B with constant velocity $(\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. Calculate the time that the particle takes to move from A to B .

(3)

- (b) The particle then moves from B to a point C with a constant acceleration of $2\mathbf{j} \text{ m s}^{-2}$. It takes 4 seconds to move from B to C .

(i) Find the position vector of C .

(4)

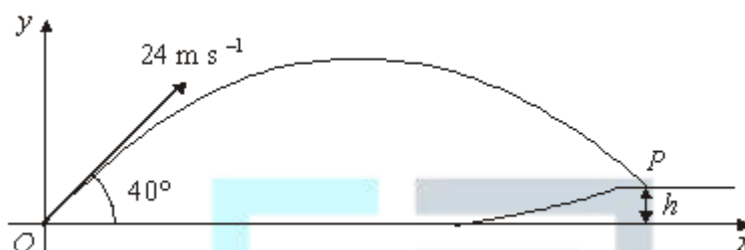
(ii) Find the distance AC .

(2)

(Total 9 marks)

Q12.

A golf ball is struck from a point O with velocity 24 m s^{-1} at an angle of 40° to the horizontal. The ball first hits the ground at a point P , which is at a height h metres above the level of O .



The horizontal distance between O and P is 57 metres.

- (a) Show that the time that the ball takes to travel from O to P is 3.10 seconds, correct to three significant figures.

(3)

(b) Find the value of h .

(3)

- (c) (i) Find the speed with which the ball hits the ground at P .

(5)

(ii) Find the angle between the direction of motion and the horizontal as the ball hits the ground at P .

(2)

(Total 13 marks)

Q13.

A stone is dropped from a high bridge and falls vertically.

- (a) Find the distance that the stone falls during the first 4 seconds of its motion.

(3)

(b) Find the average speed of the stone during the first 4 seconds of its motion.

(2)

- (c) State one modelling assumption that you have made about the forces acting on the stone during the motion.

(1)

(Total 6 marks)

Q14.

A car travels along a straight horizontal road. The motion of the car can be modelled as three separate stages.

During the first stage, the car accelerates uniformly from rest to a velocity of 10 m s^{-1} in 6 seconds.

During the second stage, the car travels with a constant velocity of 10 m s^{-1} for a further 4 seconds.

During the third stage of the motion, the car travels with a uniform retardation of magnitude 0.8 ms^{-2} until it comes to rest.

- (a) Show that the time taken for the **third** stage of the motion is 12.5 seconds.

(2)

- (b) Sketch a velocity–time graph for the car during the three stages of the motion.

(4)

- (c) Find the total distance travelled by the car during the motion.

(3)

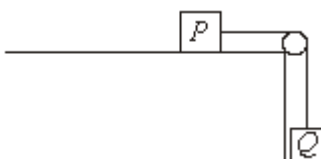
- (d) State one criticism of the model of the motion.

(1)

(Total 10 marks)

Q15.

A small block P is attached to another small block Q by a light inextensible string. The block P rests on a rough horizontal surface and the string hangs over a smooth peg so that Q hangs freely, as shown in the diagram.



The block P has mass 0.4 kg and the coefficient of friction between P and the surface is 0.5 .

The block Q has mass 0.3 kg .

The system is released from rest and Q moves vertically downwards.

- (a) (i) Draw a diagram to show the forces acting on P .

(1)

- (ii) Show that the frictional force between P and the surface has magnitude 1.96 newtons.

(2)

- (b) By forming an equation of motion for each block, show that the magnitude of the acceleration of each block is 1.4 m s^{-2} .

(5)

- (c) Find the speed of the blocks after 3 seconds of motion.

(2)

- (d) After 3 seconds of motion, the string breaks. The blocks continue to move. Comment on how the speed of each block will change in the subsequent motion. For each block, give a reason for your answer.

(4)

(Total 14 marks)

Q16.

A football is kicked from a point O on a horizontal football ground with a velocity of 20 m s^{-1} at an angle of elevation of 30° . During the motion, the horizontal and upward vertical displacements of the football from O are x metres and y metres respectively.

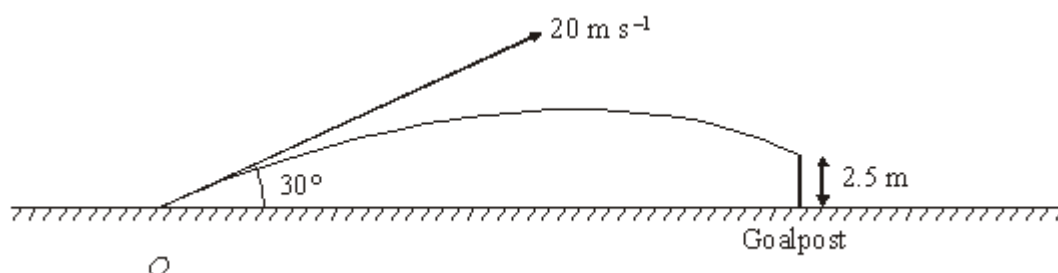
- (a) Show that x and y satisfy the equation

$$y = x \tan 30^\circ - \frac{gx^2}{800 \cos^2 30^\circ}$$

(6)

- (b) On its downward flight the ball hits the horizontal crossbar of the goal at a point which is 2.5 m above the ground. Using the equation given in part (a), find the horizontal distance from O to the goal.

(4)



- (c) State **two** modelling assumptions that you have made.

(2)

(Total 12 marks)

Q17.

A train travels along a straight horizontal track. It is travelling at a speed of 12 m s^{-1} when it begins to accelerate uniformly. It reaches a speed of 40 m s^{-1} after accelerating for 100 seconds.

- (a) (i) Show that the acceleration of the train is 0.28 m s^{-2} . (2)
- (ii) Find the distance that the train travelled in the 100 seconds. (2)
- (b) The mass of the train is 200 tonnes and a resistance force of 40 000 N acts on the train. Find the magnitude of the driving force produced by the engine that acts on the train as it accelerates. (3)
- (Total 7 marks)**

Q18.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. A yacht moves with a constant acceleration. At time t seconds the position vector of the yacht is $\mathbf{r}$ metres. When $t = 0$ the velocity of the yacht is $(2\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$, and when $t = 10$ the velocity of the yacht is $(-\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$.

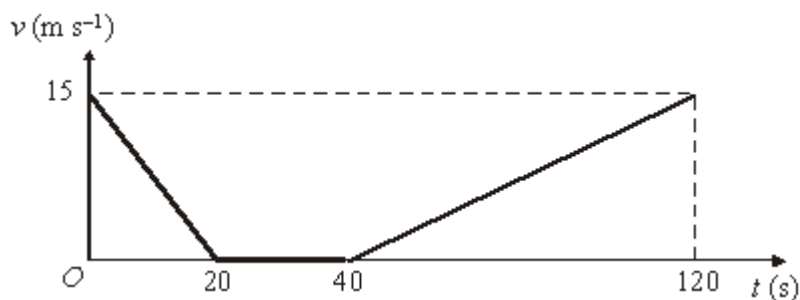
- (a) Find the acceleration of the yacht. (3)
- (b) When $t = 0$ the yacht is 20 metres due east of the origin. Find an expression for $\mathbf{r}$ in terms of t . (3)
- (c) (i) Show that when $t = 20$ the yacht is due north of the origin. (2)
- (ii) Find the speed of the yacht when $t = 20$. (4)
- (Total 12 marks)**

Q19.

A train travels along a straight horizontal track between two points A and B .

Initially the train is at A and moving at 15 m s^{-1} . Due to a problem, the train has to slow down and stop. At time $t = 40$ seconds it begins to move again. At time $t = 120$ seconds the train is at B and moving at 15 m s^{-1} again.

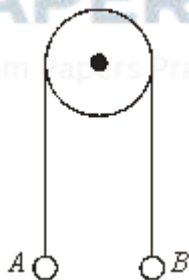
The graph below shows how the velocity of the train varies as it moves from A to B .



- (a) Use the graph to find the total distance between the points A and B . (4)
- (b) The train should have travelled between A and B at a constant velocity of 15 m s^{-1} .
- (i) Calculate the time that the train would take to travel between A and B at a speed of 15 m s^{-1} . (1)
- (ii) Calculate the time by which the train was delayed. (1)
- (c) The train has mass 500 tonnes. Find the resultant force acting on the train when $40 < t < 120$. (4)
- (Total 10 marks)

Q20.

Two particles, A of mass 5 kg and B of mass 9 kg , are attached to the ends of a light inextensible string. The string passes over a light smooth pulley as shown in the diagram. The particles are released from rest at the same height.



- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 2.8 m s^{-2} . (5)
- (b) Find the tension in the string. (2)
- (c) When B has been moving for 0.5 seconds it hits the floor. Find the height of A , above the floor, at this time. Assume that A is still below the pulley when B hits the floor.

(4)
(Total 11 marks)

Q21.

An arrow is fired horizontally with a speed of 30 m s^{-1} from a height of 2 m above ground level. Model the arrow as a particle which does not experience air resistance.

- (a) Show that the time taken for the arrow to hit the ground is 0.639 seconds, correct to three significant figures.

(4)

- (b) Find the horizontal distance travelled by the arrow.

(2)

- (c) Describe what happens, during the flight of the arrow, to the magnitude of:

- (i) the horizontal component of the velocity;
(ii) the vertical component of the velocity.

(2)

(Total 8 marks)

Q22.

A particle moves on a smooth horizontal surface with acceleration $(3\mathbf{i} - 5\mathbf{j}) \text{ m s}^{-2}$. Initially the velocity of the particle is $4\mathbf{j} \text{ m s}^{-1}$.

- (a) Find an expression for the velocity of the particle at time t seconds.

(2)

- (b) Find the time when the particle is travelling in the $\mathbf{i}$ direction.

(2)

- (c) Show that when $t = 4$ the speed of the particle is 20 m s^{-1} .

(4)

(Total 8 marks)

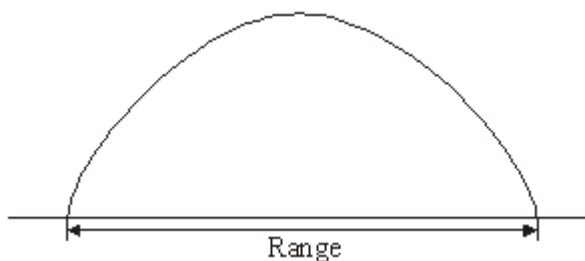
Q23.

A ball is hit from horizontal ground with velocity $(10\mathbf{i} + 24.5\mathbf{j}) \text{ m s}^{-1}$ where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertically upwards respectively.

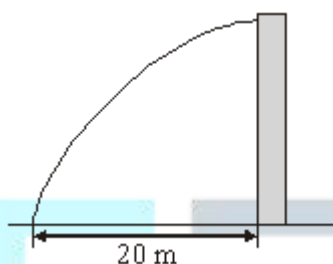
- (a) State two assumptions that you should make about the ball in order to make predictions about its motion.

(2)

- (b) The path of the ball is shown in the diagram.



- (i) Show that the time of flight of the ball is 5 seconds. (4)
- (ii) Find the range of the ball. (2)
- (c) In fact the ball hits a vertical wall that is 20 metres from the initial position of the ball.



- Find the height of the ball when it hits the wall. (4)
- (d) If a heavier ball were projected in the same way, would your answers to part (b) of this question change? Explain why. (2)

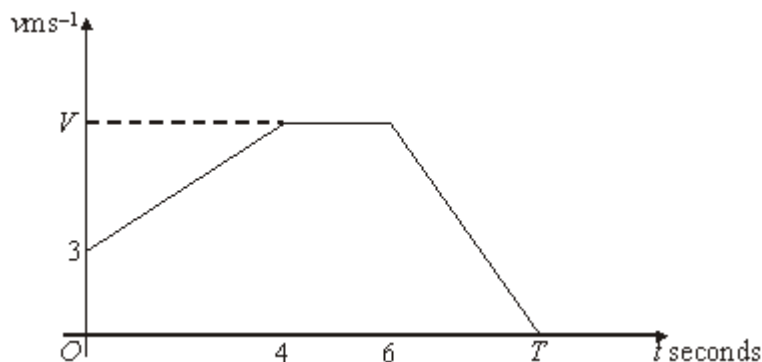
(Total 14 marks)

Q24.

The velocity–time graph below models the motion of an athlete, Paula, during part of a training exercise.

When $t = 0$ she has velocity 3 m s^{-1} . She accelerates at a constant rate until $t = 4$ and her velocity is $V \text{ m s}^{-1}$.

She runs with velocity $V \text{ m s}^{-1}$ for 2 seconds and then decelerates at a constant rate of $f \text{ m s}^{-2}$ until she comes to rest when $t = T$.



- (a) Between $t = 0$ and $t = 4$, Paula runs a distance of 20 metres. Show that $V = 7$. (2)
- (b) The total distance Paula runs between $t = 4$ and $t = T$ is 35 metres.
- (i) Find the value of T . (4)
- (ii) Find the value of f . (2)
- (c) Find Paula's average speed between $t = 0$ and $t = T$. (2)
- (Total 10 marks)

Q25.

A car moves in a straight line up a hill which is inclined at an angle of α to the horizontal, where $\sin \alpha = \frac{1}{14}$. The mass of the car is 1200 kg.

- (a) During the first part of its motion, the car moves with constant speed and is subject to a propulsive force of 1500 newtons and a resistance force of R newtons.
- (i) Draw a diagram showing the forces acting on the car. (2)
- (ii) Show that $R = 660$. (4)
- (iii) The car travels 60 metres in 4 seconds. Write down the speed of the car. (1)
- (b) During the second part of its motion, the propulsive force ceases to act and the resistance force is still 660 newtons. The car continues to move up the hill until it comes to rest.
- (i) Find the magnitude of the retardation of the car in this stage. (2)

- (ii) Find the time between the propulsive force ceasing to act and the car coming to rest.

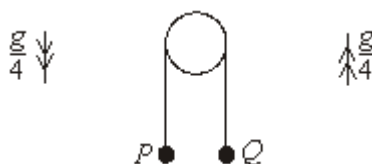
(2)

(Total 11 marks)

Q26.

Two particles, P and Q , are attached to the ends of a light inextensible string which hangs over a smooth fixed peg, as shown in the diagram. The masses of P and Q are $5m$ and km respectively, where $k < 5$.

The particles are released from rest and move with acceleration of magnitude $\frac{g}{4}$.



- (a) Find, in terms of m and g , the tension in the string.

(3)

- (b) Find the value of k .

(3)

- (c) Find the magnitude of the force on the peg due to the string.

(1)

- (d) The particles are released from rest at the same horizontal level when $t = 0$.

Find, in terms of g , the vertical distance between the particles when $t = \frac{2}{3}$.

(3)

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(Total 10 marks)

Q27.

The velocity-time graph below models the first 8 seconds of a journey of a bus.

During the first 3 seconds of its journey, the bus travels from rest with constant acceleration until it reaches a speed of $V \text{ m s}^{-1}$.

During the next 5 seconds of its journey, the bus travels with constant speed $V \text{ m s}^{-1}$.



- (a) (i) During the first 3 seconds the bus accelerates at 0.8 m s^{-2} .

Show that $V = 2.4$.

(2)

- (ii) Find the total distance the bus travels during the first 8 seconds of its journey.

(3)

- (b) After the first 8 seconds of its journey, the bus decelerates at a constant rate until it stops. During this stage of the journey the bus travels a further 4.8 metres.

Find the average speed of the bus during the whole journey.

(4)

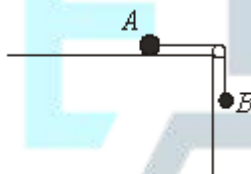
(Total 9 marks)

Q28.

A particle A of mass 0.2 kg is connected to a particle B of mass 0.3 kg by a light inextensible string.

The particle A is placed on a rough horizontal surface.

The string passes over a smooth fixed peg with B hanging freely, as shown in the diagram.



- (a) The system is held in equilibrium by a horizontal force, P newtons, which is applied to the particle A . The coefficient of friction between A and the rough horizontal surface is 0.5 .

- (i) Draw a diagram showing the forces acting on the particle A .

(2)

- (ii) Show that the magnitude of the maximum possible frictional force between A and the rough horizontal surface is 0.98 newtons.

(2)

- (iii) Find the tension in the string.

(2)

- (iv) Find the least value of P required to hold the system in equilibrium.

(2)

- (b) The horizontal force P is removed and the particles start to move from rest.

- (i) Find the magnitude of the acceleration of the particles during the subsequent motion.

(5)

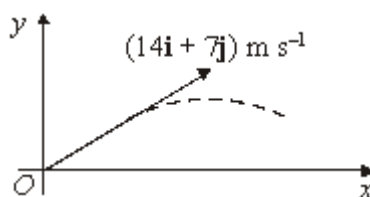
- (ii) After falling 0.1 metres, the particle B hits the ground. Find the time between the system being released and B hitting the ground.

(2)

(Total 15 marks)

Q29.

Jonny is practising kicking a rugby ball. He kicks the ball from a point O on horizontal ground with velocity $14\mathbf{i} + 7\mathbf{j}$ m s⁻¹. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in the directions of axes Ox and Oy , and Oy is vertically upwards.



- (a) (i) Find the coordinates of the ball t seconds after it has been kicked.

(3)

- (ii) Show that the equation of the path of the ball is

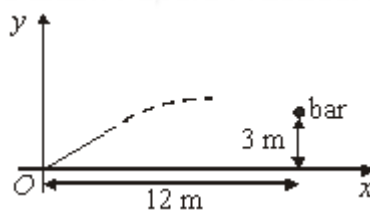
$$y = \frac{20x - x^2}{40}$$

(3)

- (iii) Find the horizontal range of the ball.

(2)

- (b) The ball moves towards a horizontal bar which is 3 metres above the ground and at a horizontal distance of 12 metres from O , as shown in the diagram.



Determine whether the ball passes above or below the bar.

(3)

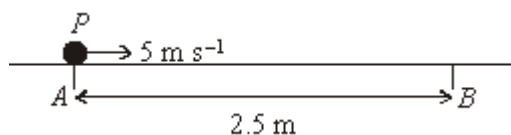
- (c) State one modelling assumption that you have used in answering this question.

(1)

(Total 12 marks)

Q30.

A particle P moves in a straight line across a horizontal surface with retardation of magnitude 1.8 m s^{-2} . The particle P passes through a point with velocity 5 m s^{-1} , as shown in the diagram. It subsequently passes **twice** through the point B , where $AB = 2.5$ metres.

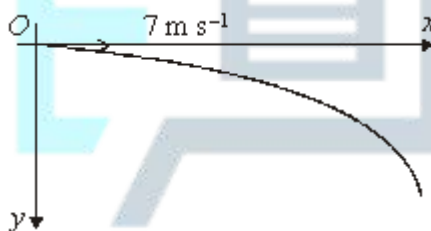


- (a) Find the velocities of P on the two occasions when it passes through B . (3)
- (b) Hence, or otherwise, find the length of time between the two occasions when P passes through B . (3)

(Total 6 marks)

Q31.

Elaine kicks a ball off a cliff top. She kicks the ball from a point O and it subsequently moves in a vertical plane with respect to axes which are horizontal and vertically downwards, as shown in the diagram.



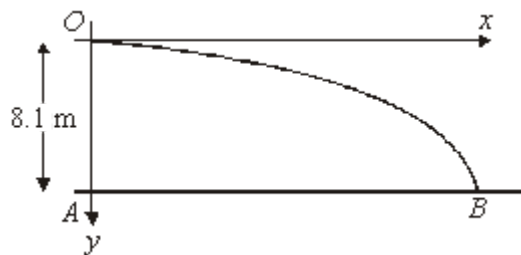
The initial velocity of the ball is horizontal and of magnitude 7 m s^{-1} .

- (a) Find the coordinates of the ball t seconds after it has been kicked. (3)
- (b) Show that the equation of the path of the ball is

$$y = \frac{x^2}{10}.$$

(2)

The ball subsequently lands on a horizontal beach. The point A on the beach is vertically below O and the ball lands at the point B on the beach, as shown in the diagram.



- (c) The distance OA is 8.1 metres.

Find the distance AB .

(2)

- (d) Find the speed of the ball as it reaches B .

(5)

(Total 12 marks)

Q32.

A child kicks a small toy brick in a straight line across a horizontal floor. The brick initially moves at 3.5 m s^{-1} and comes to rest in a distance of 2.5 metres.

- (a) Show that the magnitude of the retardation of the brick is 2.45 m s^{-2} .

(3)

- (b) The mass of the brick is 0.2 kg.

- (i) Find the magnitude of the frictional force acting on the brick.

(2)

- (ii) Find the coefficient of friction between the brick and the floor.

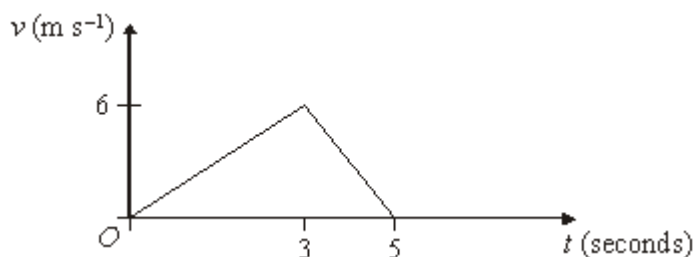
(2)

(Total 7 marks)

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Q33.

The velocity–time graph below models Tom's motion as he runs across a playground.



- (a) Find Tom's acceleration during the first 3 seconds of his motion.

(2)

- (b) Find the total distance Tom runs during the 5 seconds of motion.

(3)

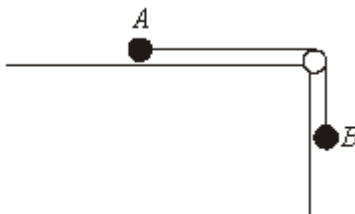
- (c) State one criticism of this model of Tom's motion.

(1)

(Total 6 marks)

Q34.

Two particles, A and B , are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



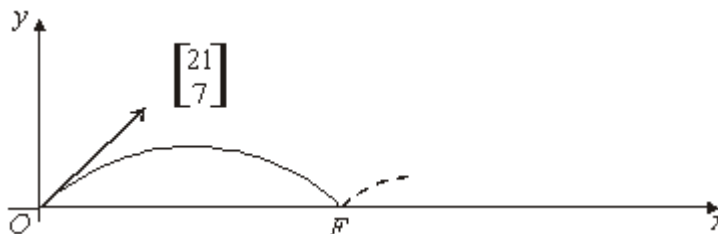
The particle A is of mass 0.5 kg and the particle B is of mass 0.1 kg. The particle A is in contact with a rough horizontal surface.

- (a) The system is at rest and the particle B hangs vertically.
- (i) Show that the tension in the string is 0.98 newtons. (1)
- (ii) The particle A rests in limiting equilibrium. Show that the coefficient of friction between A and the surface is 0.2. (3)
- (iii) Draw a diagram to show the forces acting on the peg due to the string. (1)
- (iv) Find the magnitude of the resultant force on the peg due to the string. (2)
- (b) An additional mass of 0.1 kg is attached to the particle B and the system is released from rest. The particle A moves across the surface towards the peg and B moves vertically downwards. The particles move with acceleration $a \text{ m s}^{-2}$ and the tension in the string is T newtons.
- (i) Show that $a = 1.4$. (5)
- (ii) Find the value of T . (1)
- (iii) After falling a distance of 0.7 metres, B hits the ground. Find the time between the particles being released and B hitting the ground. (2)

(Total 15 marks)

Q35.

A ball is kicked from a point O on horizontal ground with velocity $\begin{bmatrix} 21 \\ 7 \end{bmatrix} \text{ m s}^{-1}$. The ball subsequently hits the ground at a point F , as shown in the diagram.



- (a) (i) Find the time between the ball being kicked and reaching F .

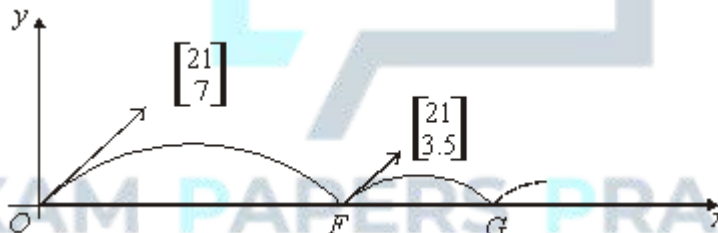
(3)

- (ii) Find the distance OF .

(2)

- (b) The ball rebounds from F with velocity $\begin{bmatrix} 21 \\ 3.5 \end{bmatrix} \text{ m s}^{-1}$. It subsequently hits the ground again at

G and then rebounds, as shown in the diagram below.



Find the distance FG .

(4)

- (c) After bouncing at G , the ball rebounds with velocity $\begin{bmatrix} 21 \\ 1.75 \end{bmatrix} \text{ m s}^{-1}$, so that the vertical component of the velocity is halved again, and then subsequently bounces next at H .

Find the total distance OH .

(3)

(Total 12 marks)